

HBTU Chemistry Unit 5 Study Guide

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User prompt: Polymers: Introduction, types of polymers, Molecular mass-number and mass average molecular weight, determination of molecular mass by Osmometry, viscosity, light scattering and size exclusion chromatography, Rheological properties and uses of some common polymers. Synthetic Polymers (carbon framework, silicon framework, fluorinated polymer), Conducting and Biodegradable polymers. Water Analysis: Introduction; Hardness of Water- cause, types, units, Disadvantages of using hard water for domestic and industrial purposes, Softening of hard water, Chemical analysis of Water- estimation of free chlorine, total alkalinity, hardness, Numerical based on determination of hardness. Solid Waste Management: Classification, waste treatment & Disposal methods (Composting, sanitary landfilling, thermal processes, recycling and reuse Ye unit 5 hai Hbtu chem all topics should be cover without leaving any single topics give me best pdf for given topic Without leaving anything Give me PDF

Response: Engineering Chemistry (BCY-151/152) – Module V: Master Study Guide

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Part 1: Polymers

1.1 Introduction and Basic Concepts

A **polymer** is a high molecular weight macromolecule formed by the covalent linking of repeating structural units known as **monomers**.

- **Polymerization:** The chemical process of linking monomer molecules together in a chain or three-dimensional network.
- **Monomer:** A small, reactive simple molecule containing multiple bonding sites or at least two functional groups capable of forming covalent bonds with other monomers.
- **Functionality:** The number of reactive sites or bonding positions present in a monomer molecule.
 - *Bifunctional* monomers (e.g., ethylene, $\text{H}_2\text{C} = \text{CH}_2$; adipic acid) yield linear chains.
 - *Trifunctional or polyfunctional* monomers (e.g., glycerol, phenol) yield branched or three-dimensional network structures.
- **Degree of Polymerization (DP):** The total number of repeating monomeric units present in a polymer macromolecule:

$$DP = \frac{M}{\bar{m}_0}$$

where M is the molecular mass of the polymer chain and $\bar{m}_0$ is the molecular mass of the repeating unit.

1.2 Classification of Polymers

By Origin

- **Natural Polymers:** Isolated directly from plants or animals (e.g., cellulose, starch, natural rubber, proteins, DNA).
- **Semi-synthetic Polymers:** Chemically modified natural polymers (e.g., cellulose acetate / rayon, vulcanized rubber, nitrocellulose).
- **Synthetic Polymers:** Man-made laboratory/industrial polymers synthesized from petro-derivatives (e.g., polyethylene, nylon-6,6, polystyrene, Bakelite).

By Structural Framework

- **Linear Polymers:** Monomers join head-to-tail in long continuous chains. Chains pack tightly, leading to high density, high melting point, and high tensile strength (e.g., High-Density Polyethylene [HDPE], PVC).
- **Branched-chain Polymers:** Main polymer backbones possess short or long pendant branches. Packing efficiency drops, resulting in lower density and lower melting point (e.g., Low-Density Polyethylene [LDPE], glycogen).
- **Cross-linked / Network Polymers:** Bifunctional and polyfunctional monomers form transverse covalent cross-links between chains, yielding rigid 3D structures (e.g., Bakelite, Melamine-formaldehyde resin).

By Polymerization Mechanism

- **Addition / Chain-Growth Polymerization:** Monomers containing double or triple bonds link without the loss of any small molecular by-products. The empirical formula of the polymer matches that of the monomer (e.g., Polyethylene, Polystyrene, Teflon). Operates via Free Radical, Cationic, or Anionic mechanisms.
- **Condensation / Step-Growth Polymerization:** Bifunctional or polyfunctional monomers react step-by-step with the elimination of small side products such as H_2O , HCl , NH_3 , or CH_3OH (e.g., Nylon-6,6, Terylene / Dacron, Bakelite).

By Intermolecular Forces

Class	Characteristic Intermolecular Forces	Mechanical Properties	Typical Examples
Elastomers	Very weak van der Waals forces	High elasticity, stretches under tension and returns to shape	Natural rubber, Buna-S, Buna-N
Fibres	Strong hydrogen bonding or dipole-dipole attractions	High tensile strength, high modulus, low elongation	Nylon-6,6, Terylene, Kevlar
Thermoplastics	Intermediate forces; chains slide upon heating	Soften on heating, harden on cooling; easily remolded and recycled	Polythene, Polyvinyl chloride (PVC), Polystyrene
Thermosetting Plastics	Extensive covalent cross-linking across chains	Do not soften on heating; undergo irreversible chemical degradation; cannot be remolded	Bakelite, Urea-formaldehyde resin, Epoxy resins

By Tacticity (Stereochemistry)

- **Isotactic:** All functional substituents (−R) are aligned on the **same side** of the carbon chain backbone. Highly crystalline.
- **Syndiotactic:** Functional substituents (−R) alternate regularly on **opposite sides** along the backbone.
- **Atactic:** Substituents (−R) are dispersed in a completely **random** spatial arrangement. Typically amorphous and soft.

1.3 Molecular Mass of Polymers

Unlike simple small molecules with precise molecular weights, synthetic polymers are **polydisperse**—they consist of chains of varying lengths and molecular weights. Therefore, polymer molecular weight is expressed statistically as an average.

Number Average Molecular Weight ($\bar{M}_n$)

$\bar{M}_n$ represents the total weight of all molecules in a polymer sample divided by the total number of molecules. It is sensitive to the total number of particles (colligative properties) rather than their individual size:

$$\bar{M}_n = \frac{\sum N_i M_i}{\sum N_i}$$

where N_i is the number of polymer chains of molecular mass M_i .

Weight (Mass) Average Molecular Weight ($\bar{M}_w$)

$\bar{M}_w$ accounts for the mass fraction of each species, giving higher weight to heavier, larger macromolecules. It is determined by methods sensitive to molecular size and shape (e.g., light scattering):

$$\bar{M}_w = \frac{\sum N_i M_i^2}{\sum N_i M_i} = \sum w_i M_i$$

where $w_i = \frac{N_i M_i}{\sum N_i M_i}$ is the weight fraction of chains with molecular mass M_i .

Polydispersity Index (PDI)

The breadth of the molecular weight distribution is quantified by the Polydispersity Index:

$$PDI = \frac{\bar{M}_w}{\bar{M}_n}$$

- For a strictly **monodisperse** polymer (all chains identical in length, such as natural proteins), $\bar{M}_w = \bar{M}_n$, meaning **PDI = 1**.
- For **polydisperse** synthetic polymers, $\bar{M}_w > \bar{M}_n$, meaning **PDI > 1** (typically 1.5 to 30).

1.4 Determination of Polymer Molecular Mass

SUMMARY OF EXPERIMENTAL METHODS			
Method	Average Determined	Fundamental Phenomenon	
Osmometry (Membrane)	Number Avg ($\bar{M}_n$)	Osmotic pressure (Colligative)	
Viscometry	Viscosity Avg ($\bar{M}_v$)	Solution hydrodynamic drag	
Light Scattering	Weight Avg ($\bar{M}_w$)	Rayleigh / Mie scattering	
Size Exclusion (SEC)	Full distribution	Porous bead fractionation	

1. Osmometry (Membrane & Vapor Pressure)

Osmometry measures the osmotic pressure (π) established across a semipermeable membrane that allows solvent molecules to pass while blocking larger polymer solute chains.

- According to the **Van 't Hoff equation** adapted for non-ideal polymer solutions:

$$\frac{\pi}{c} = RT \left(\frac{1}{\bar{M}_n} + A_2c + A_3c^2 + \dots \right)$$

where c is the polymer concentration in g/L, R is the universal gas constant, T is absolute temperature in K, and A_2 is the second virial coefficient representing polymer-solvent interaction.

- Working Principle:** A series of dilute solutions of known concentration (c) are prepared. π/c is plotted against c and extrapolated linearly to zero concentration ($c \rightarrow 0$):

$$\lim_{c \rightarrow 0} \left(\frac{\pi}{c} \right) = \frac{RT}{\bar{M}_n}$$

- Limitation:** Membrane osmometry is reliable for $\bar{M}_n$ between 20,000 and 1,000,000 g/mol. Below 20,000, chains diffuse across the membrane; above 1,000,000, osmotic pressure is too minute to record accurately.

2. Viscometry

Viscometry relies on the principle that dissolving a high-molecular-weight polymer increases the viscosity of the pure solvent by disrupting streamlines during laminar flow.

- Viscosity Definitions:**

- Relative viscosity: $\eta_r = \frac{\eta}{\eta_0} \approx \frac{t}{t_0}$ (efflux time ratio in an Ostwald or Ubbelohde capillary viscometer).
- Specific viscosity: $\eta_{sp} = \eta_r - 1 = \frac{t - t_0}{t_0}$.
- Reduced viscosity: $\eta_{red} = \frac{\eta_{sp}}{c}$.
- Intrinsic viscosity $[\eta]$: Extrapolated reduced viscosity at infinite dilution:

$$[\eta] = \lim_{c \rightarrow 0} \left(\frac{\eta_{sp}}{c} \right)$$

- Mark-Houwink-Sakurada Equation:**

$$[\eta] = K \bar{M}_v^a$$

Taking the natural logarithm:

$$\ln[\eta] = \ln K + a \ln \bar{M}_v$$

where K and a are empirical constants specific to a given polymer-solvent system at a set temperature (constant a ranges between 0.5 for θ -solvents and 0.8 for good solvents). Viscometry yields the **viscosity-average molecular weight** ($\bar{M}_v$), where $\bar{M}_n < \bar{M}_v < \bar{M}_w$.

3. Light Scattering

When a collimated, polarized light beam passes through a polymer solution, the electric field induces oscillating dipoles in the polymer molecules, causing light to scatter in all directions (Rayleigh scattering).

- The scattered light intensity at angle θ (Rayleigh ratio R_θ) is directly proportional to the molecular weight of the solute and the square of the refractive index increment (dn/dc):

$$\frac{K^*c}{R_\theta} = \frac{1}{\bar{M}_w P(\theta)} + 2A_2c$$

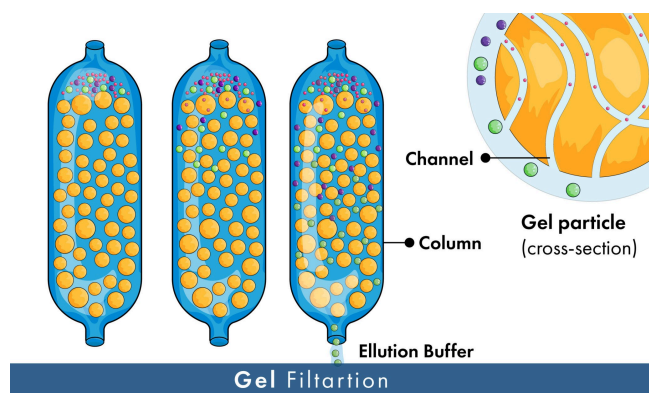
where $K^* = \frac{4\pi^2 n_0^2 (dn/dc)^2}{NA\lambda_0^4}$ is an optical constant, and $P(\theta)$ is a particle scattering function accounting for internal destructive interference.

- By measuring scattering over varying concentrations (c) and angles (θ), a two-dimensional grid called a **Zimm Plot** is constructed. Extrapolating simultaneously to $c \rightarrow 0$ and $\theta \rightarrow 0$ yields:

- The vertical intercept = $\frac{1}{\bar{M}_w}$ (yielding true weight-average molecular mass).
- The initial slope of the angular extrapolation = Radius of gyration (R_g , measuring chain dimensions).
- The initial slope of the concentration extrapolation = Second virial coefficient (A_2).

4. Size Exclusion Chromatography (SEC / GPC)

Gel Permeation Chromatography (GPC) / Size Exclusion Chromatography (SEC) separates polymer molecules strictly according to their **hydrodynamic volume** (effective size in solution).



- **Stationary Phase:** Rigid, microporous cross-linked polymer beads (e.g., styrene-divinylbenzene copolymer) with controlled pore size distributions.
- **Mobile Phase:** High-purity organic solvent (e.g., Tetrahydrofuran [THF], chloroform) or aqueous buffer running under high pressure.
- **Mechanism:**
 - **Large polymer macromolecules** with a hydrodynamic radius larger than the pore cut-off cannot enter the pores and are completely excluded. They travel exclusively through the interstitial void volume (V_0) and **elute first**.
 - **Small polymer molecules and oligomers** penetrate deep into the tortuous pore network of the beads. They travel through an expanded effective volume ($V_0 + V_i$) and **elute last**.
- **Elution Volume Equation:**

$$V_e = V_0 + K_{SEC} V_i$$

where K_{SEC} is the partition coefficient ($0 \leq K_{SEC} \leq 1$).

- Calibrating with monodisperse standards (such as polystyrene) relates retention volume to $\log M$, producing the entire molecular weight distribution curve ($\bar{M}_n$, $\bar{M}_w$, and PDI).

1.5 Rheological Properties of Polymers

Rheology examines the flow, deformation, and time-dependent mechanical response of polymeric melts and solutions.

- **Viscoelasticity:** Polymers simultaneously exhibit viscous liquid characteristics (permanent energy dissipation via chain slipping) and elastic solid characteristics (reversible energy storage through uncoiling of molecular chains).
- **Glass Transition Temperature (T_g) vs. Melting Temperature (T_m):**
 - T_g : The temperature at which an amorphous polymer transitions from a hard, brittle, glassy state to a flexible, rubbery state. It marks the onset of coordinated segmental motion of the polymer backbone (typically 20–50 carbon atoms).
 - T_m : The thermodynamic first-order phase transition temperature where crystalline domains in semi-crystalline polymers melt into a disordered liquid.
- **Factors Influencing T_g :**
 - **Chain flexibility:** Flexible backbones (e.g., $-\text{Si}-\text{O}-$, $-\text{C}-\text{O}-$) yield lower T_g ; rigid aromatic backbones elevate T_g .
 - **Bulky pendant groups:** Bulky rings (e.g., polystyrene, $T_g \approx 100^\circ\text{C}$) restrict rotation, increasing T_g .
 - **Intermolecular forces:** Polar groups and hydrogen bonding elevate T_g (e.g., Nylon-6,6).
 - **Cross-linking:** High cross-linking density severely limits segmental motion, driving T_g upward.
 - **Plasticizers:** Low molecular weight additives increase free volume between chains, significantly lowering T_g .
- **Non-Newtonian Flow:** Polymer melts display **pseudoplasticity (shear-thinning)**: as shear rate ($\dot{\gamma}$) increases, viscosity (η) drops because polymer chains uncoil, disentangle, and align along the streamline directions.

1.6 Synthetic Polymers & Their Uses

1. Carbon Framework Polymers

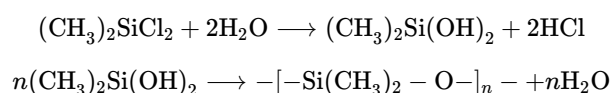
- **Polyethylene (PE):**

- **Low-Density Polyethylene (LDPE):** Highly branched structure; synthesized at high pressure (1000–3000 atm, 200–300 °C) via free radical mechanism. Flexible, low density (0.91–0.94 g/cm³). Used for packaging films, squeeze bottles, and wire insulation.
- **High-Density Polyethylene (HDPE):** Linear, unbranched; synthesized at low pressure (2–6 atm, 60–80 °C) using Ziegler-Natta catalysts (TiCl₄ + Al(C₂H₅)₃). Highly crystalline, high density (0.94–0.97 g/cm³). Used for pipes, fuel tanks, and heavy-duty containers.
- **Polyvinyl Chloride (PVC):**
 - Monomer: Vinyl chloride (CH₂ = CHCl).
 - Rigid and chemically resistant. Compounded with dioctyl phthalate (DOP) plasticizers for flexibility. Used in construction conduits, plumbing pipes, raincoats, and blood bags.
- **Bakelite (Phenol-Formaldehyde Resin):**
 - Thermosetting polymer synthesized via step-growth condensation of phenol with formaldehyde in the presence of an acid or base catalyst.
 - Step 1 produces linear *Novolac* under acidic conditions.
 - Step 2 involves curing with hexamethylenetetramine under heat to produce a highly cross-linked, rigid, insoluble 3D network.
 - Used for electrical switches, plugs, printed circuit boards (PCBs), and frying pan handles.

2. Silicon Framework Polymers (Silicones / Polysiloxanes)

Inorganic-organic hybrid polymers with an alternating silicon-oxygen backbone (–Si – O – Si–) and organic pendant groups (usually methyl, –CH₃):

- **Synthesis:** Hydrolysis of dimethyldichlorosilane yields silanols that condense into linear polydimethylsiloxane (PDMS):



Adding methyltrichlorosilane (CH₃SiCl₃) introduces 3D cross-linking. Trimethylchlorosilane ((CH₃)₃SiCl) acts as a chain terminator to control molecular weight.

- **Properties:** High thermal stability (–60 °C to +250 °C), water repellency (hydrophobic methyl sheath), low surface tension, excellent electrical insulation, and chemical inertness.
- **Uses:** High-temperature lubricants, waterproof sealants, biocompatible medical implants, catheter tubing, and flexible cookware.

3. Fluorinated Polymers (Teflon / PTFE)

- **Structure & Synthesis:** Polytetrafluoroethylene (PTFE) is synthesized by free-radical emulsion polymerization of tetrafluoroethylene (CF₂ = CF₂) under pressure with ammonium persulfate initiator:



- **Properties:** Exceptional thermal stability (up to 260 °C continuous service), universal chemical inertness (the dense fluorine electron cloud shields the C – C backbone, and the C – F bond energy is extremely high at 485 kJ/mol), lowest coefficient of friction of any known solid, non-wetting.
- **Uses:** Non-stick cookware coatings, gaskets and seals for corrosive chemical service, high-frequency coaxial cable insulation, Gore-Tex breathable membranes.

1.7 Conducting Polymers

Intrinsically Conducting Polymers (ICPs) are organic macromolecules with extensive **conjugated alternating single and double bonds** (π-electron delocalization) along their backbones.

- **Mechanism of Conduction:** The continuous overlap of carbon *p_z* orbitals forms delocalized π (valence) and π* (conduction) bands with an initial energy band gap of 1.5 to 3 eV (semiconducting or insulating in the pristine state).
- **Doping Process:** Electrical conductivity increases by several orders of magnitude (up to metallic levels, ≈ 10³ to 10⁵ S/cm) via chemical or electrochemical oxidation/reduction:
 - **p-doping (Oxidation):** Polymer backbone is treated with an oxidizing agent (e.g., I₂, FeCl₃, AsF₅). An electron is extracted from the valence band, forming a radical cation (**polaron**). Further oxidation removes another electron, generating a dication (**bipolaron**). These mobile charge carriers migrate under an electric field.
 - **n-doping (Reduction):** Polymer backbone is treated with a strong reducing agent (e.g., Sodium naphthalide), adding electrons to the conduction band.
- **Common Examples:**
 - *Polyaniline (PANI)* (emeraldine salt form displays high conductivity).

- *Polyacetylene* ($[-CH=CH-]_n$).

- *Polypyrrole (PPy)*, *Polythiophene (PT)*.

- **Applications:** Antistatic packaging, electrochromic smart windows, organic light-emitting diodes (OLEDs), biosensors, lightweight flexible battery electrodes, electromagnetic shielding.

1.8 Biodegradable Polymers

Polymers that undergo microbially mediated chain cleavage (by bacteria, fungi, or enzymes) into harmless mineral end-products: CO_2 , H_2O , and biomass under aerobic conditions, or CH_4 , CO_2 , and biomass under anaerobic conditions.

Degradation Mechanism

Enzymatic hydrolysis or microbial oxidation targets hydrolyzable linkages along the polymer backbone:

- Ester bonds ($-COO-$)
- Amide bonds ($-CONH-$)
- Glycosidic bonds

Chain scission lowers molecular weight below 5000 g/mol, enabling cellular assimilation by microorganisms.

Major Examples

1. PHBV (Polyhydroxybutyrate-co- β -hydroxyvalerate):

- Synthesized by bacterial fermentation of glucose with *Ralstonia eutropha*.
- Copolymer of 3-hydroxybutanoic acid and 3-hydroxypentanoic acid.
- *Properties:* Semi-crystalline, thermoplastic, degrades cleanly without toxic residue.
- *Uses:* Controlled drug delivery capsules, orthopedic pins, specialty packaging.

2. Polylactic Acid (PLA):

- Produced by ring-opening polymerization of lactide dimers derived from fermented corn starch.
- *Uses:* 3D printing filaments, disposable food packaging containers, compostable bottles.

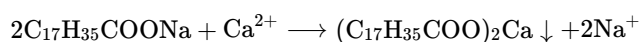
3. PGA (Polyglycolic Acid):

- High-strength bioresorbable aliphatic polyester.
- *Uses:* Absorbable internal surgical sutures (Dexon), internal bone fixators.

Part 2: Water Analysis

2.1 Introduction and Hardness of Water

Hardness is the soap-consuming capacity of a water sample. Hard water precipitates insoluble curd-like salts when shaken with soap (sodium or potassium stearate):



Causes of Hardness

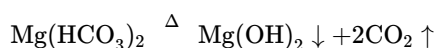
Hardness is caused by polyvalent multivalent metal cations dissolved in water, primarily Ca^{2+} and Mg^{2+} , and to a lesser extent Fe^{2+} , Mn^{2+} , and Al^{3+} . Natural dissolved CO_2 leaches these ions from limestone ($CaCO_3$) and dolomite ($CaCO_3 \cdot MgCO_3$):



Types of Hardness

1. Temporary Hardness (Carbonate Hardness):

- Caused by dissolved **bicarbonates of calcium and magnesium**: $Ca(HCO_3)_2$ and $Mg(HCO_3)_2$.
- Readily removed by **boiling**, which decomposes soluble bicarbonates into insoluble precipitates:



2. Permanent Hardness (Non-Carbonate Hardness):

- Caused by dissolved **chlorides, sulfates, and nitrates of calcium and magnesium**: CaCl_2 , MgCl_2 , CaSO_4 , MgSO_4 .
- Cannot be removed by boiling; requires specialized chemical softening techniques.

$$\text{Total Hardness} = \text{Temporary Hardness} + \text{Permanent Hardness}$$

Units of Hardness

Hardness is universally expressed in terms of CaCO_3 **equivalents**:

$$\text{Hardness in terms of } \text{CaCO}_3 \text{ eq} = \frac{\text{Mass of hardness-causing substance}}{\text{Equivalent weight of substance}} \times \text{Equivalent weight of } \text{CaCO}_3 \text{ (50)}$$

Why CaCO_3 is used as the standard:

1. Its molecular weight is exactly 100 g/mol, and equivalent weight is 50 g/eq, making calculations direct and simple.
2. It is the most insoluble precipitate formed during chemical water-softening reactions.

Common Units:

- **ppm (Parts per million)**: Parts of CaCO_3 equivalent hardness per 10^6 parts of water.
- **mg/L**: Milligrams of CaCO_3 equivalent hardness per liter of water (1 L of water $\approx 10^6$ mg, hence 1 ppm = 1 mg/L).
- **Degree French ($^\circ\text{Fr}$)**: Parts of CaCO_3 equivalent hardness per 10^5 parts of water (1 $^\circ\text{Fr}$ = 10 ppm).
- **Degree Clark ($^\circ\text{Cl}$)**: Grains of CaCO_3 equivalent hardness per imperial gallon (70,000 grains) of water (1 $^\circ\text{Cl}$ = 14.3 ppm).

$$1 \text{ ppm} = 1 \text{ mg/L} = 0.1 \text{ }^\circ\text{Fr} = 0.07 \text{ }^\circ\text{Cl}$$

Equivalent Weights of Common Species:

- $\text{Ca}(\text{HCO}_3)_2$: 81
- $\text{Mg}(\text{HCO}_3)_2$: 73
- CaSO_4 : 68
- CaCl_2 : 55.5
- MgSO_4 : 60
- MgCl_2 : 47.5
- Ca^{2+} : 20
- Mg^{2+} : 12.2
- HCO_3^- : 61
- CO_3^{2-} : 30
- OH^- : 17

2.2 Disadvantages of Using Hard Water

Domestic Disadvantages

- **Soap Consumption**: Wastes large amounts of soap before lather forms; insoluble curds adhere to fabrics, causing discoloration and stiff textures.
- **Cooking**: Elevates the boiling point of water, lengthening cooking times; leaves mineral deposits on vegetables and tea infusions.
- **Health & Plumbing**: Causes skin irritation; forms mineral scale inside domestic geysers, showerheads, and municipal supply pipes, restricting flow.

Industrial Disadvantages (Boiler Troubles)

Using hard feed-water in high-pressure steam boilers leads to four major operating failures:

BOILER TROUBLES SUMMARY

Trouble	Primary Causes & Consequences
Scales & Sludges	CaSO_4 , silicates / MgCO_3 , MgSO_4 ; fuel waste, tube burst
Boiler Corrosion	Dissolved O_2 , CO_2 , MgCl_2 hydrolysis; wall thinning

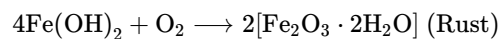
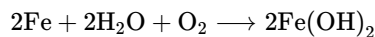
Priming & Foaming	High dissolved solids, oils/fats; wet steam carried over
Caustic Embrittlement	Free NaOH in alkaline water; intergranular micro-cracks

1. Scale and Sludge Formation:

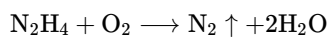
- **Sludge:** Soft, loose, non-adherent precipitates formed in colder parts of the boiler by salts with retrograde solubility (e.g., MgCO_3 , MgCl_2 , MgSO_4). Removed easily via periodic **blowdown operations**.
- **Scale:** Hard, dense, adherent crystalline coatings deposited directly on boiler heating surfaces (e.g., CaSO_4 , CaSiO_3 , Ca(OH)_2).
- **Hazards:** Scales are poor thermal conductors. Overheating boiler steel tubes to force heat transfer leads to structural failure, pipe bulging, and explosions.

2. Boiler Corrosion:

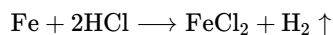
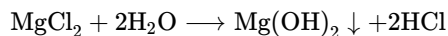
- *Dissolved O_2 :* At elevated temperatures:



Mitigation: Add chemical deoxygenators (Na_2SO_3 or hydrazine, N_2H_4):



- *Dissolved CO_2 :* Forms carbonic acid (H_2CO_3). Neutralized by adding ammonium hydroxide (NH_4OH).
- *Hydrolyzable salts:* Salts like MgCl_2 hydrolyze at boiler temperatures to release free HCl , which attacks iron surfaces:

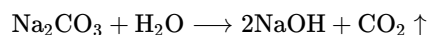


3. Priming and Foaming (Carry-Over):

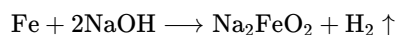
- *Priming:* Rapid boiling projects fine water droplets into escaping steam, producing "wet steam" that deposits salts inside superheaters and erodes steam turbine blades.
- *Foaming:* Formation of persistent bubbles on the surface of boiler water, caused by high concentrations of suspended solids, oils, or organic surfactants.

4. Caustic Embrittlement:

- A form of stress corrosion cracking in high-pressure boilers. Alkaline feed-water containing sodium carbonate hydrolyzes:



- Concentrated NaOH enters hairline crevices and stressed rivet joints via capillary action. It dissolves protective iron oxides, forming sodium ferroate:

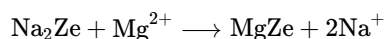
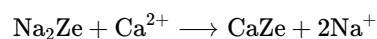


- This leaves the metal brittle along grain boundaries, eventually leading to sudden mechanical failure.

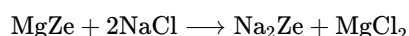
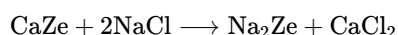
2.3 Softening of Hard Water

1. Zeolite (Permutit) Process

- **Chemical Structure:** Zeolites are hydrated sodium aluminosilicates: $\text{Na}_2\text{O} \cdot \text{Al}_2\text{O}_3 \cdot x\text{SiO}_2 \cdot y\text{H}_2\text{O}$ (abbreviated as Na_2Ze).
- **Softening Mechanism:** Water trickles through a packed bed of zeolite. Sodium ions in the zeolite matrix exchange with incoming divalent Ca^{2+} and Mg^{2+} ions:



- **Regeneration:** Once the bed is exhausted, its exchange capacity is restored by backwashing with a 10% brine (NaCl) solution:



- **Advantages:** Residual hardness is reduced to < 15 ppm; process is clean and compact; operation is straightforward.
- **Limitations:** Does not eliminate total dissolved solids (TDS); raw water containing high turbidity or dissolved $\text{Fe}^{2+}/\text{Mn}^{2+}$ fouls the zeolite matrix.

2. Ion-Exchange (Demineralization / Deionization) Process

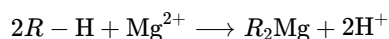
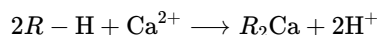
Produces high-purity demineralized water (< 2 ppm hardness, electrical conductivity $< 0.1 \mu\text{S}/\text{cm}$) using synthetic polymeric resin beads.

RAW WATER $\rightarrow$ [Cation Bed (R-H)] $\rightarrow$ [Anion Bed (R'-OH)] $\rightarrow$ [Degasifier (CO₂)] $\rightarrow$ DEMIN WATER

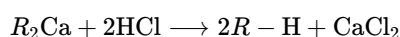
Removes: Ca, Mg, Na	Removes: Cl, SO ₄ , HCO ₃	Removes: H ₂ CO ₃ $\rightarrow$ CO ₂ + H ₂ O
Releases: H ⁺	Releases: OH ⁻	
(H ⁺ + OH ⁻ $\rightarrow$ H ₂ O)		

1. Cation Exchange Column (R – H⁺):

- High-capacity cross-linked polystyrene-divinylbenzene resin with sulfonic acid functional groups ($-\text{SO}_3^-\text{H}^+$).
- Cations exchange with resin H^+ :

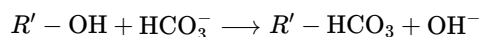
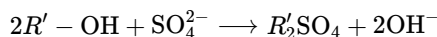
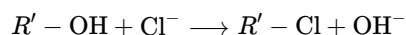


- Regeneration:* Flushed with dilute mineral acid (2–5% HCl or H₂SO₄):



2. Anion Exchange Column (R' – OH⁻):

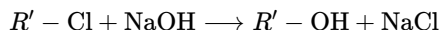
- Styrene-divinylbenzene resin functionalized with quaternary ammonium groups ($-\text{N}^+(\text{CH}_3)_3\text{OH}^-$).
- Anions exchange with resin OH^- :



- Released H^+ and OH^- combine to form pure water:



- Regeneration:* Flushed with dilute caustic soda (2–5% NaOH):

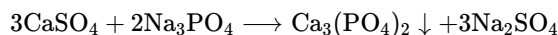


3. **Degasifier:** Air is blown upward through a packed tower to strip dissolved carbonic acid ($\text{H}_2\text{CO}_3 \rightarrow \text{H}_2\text{O} + \text{CO}_2 \uparrow$).

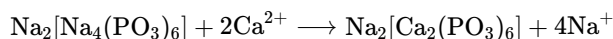
3. Internal Conditioning Methods

Added directly to water inside operating boilers:

- Phosphate Conditioning:** Adding Na_3PO_4 at high boiler pH (> 10) precipitates non-adherent calcium phosphate sludge:



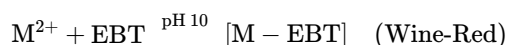
- Calgon Conditioning:** Adding sodium hexametaphosphate ($(\text{NaPO}_3)_6$ or Calgon) forms a highly soluble complex with calcium ions, keeping scale out of solution:



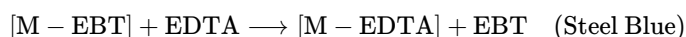
2.4 Chemical Analysis of Water

1. Estimation of Hardness by EDTA Method

- Principle:** Disodium ethylenediaminetetraacetate ($\text{Na}_2\text{H}_2\text{Y}$) forms stable, soluble 1 : 1 chelate complexes with Ca^{2+} and Mg^{2+} at an alkaline pH of 9.0–10.0 maintained using an ammoniacal buffer ($\text{NH}_4\text{Cl} + \text{NH}_4\text{OH}$).
- Indicator:** Eriochrome Black T (EBT). EBT initially binds metal ions to form a weak, wine-red complex:



- Titration:** Titrated with standard EDTA solution. EDTA has a higher binding affinity for metal ions than EBT, displacing it at the equivalence point:



The sharp color change from **wine-red to steel blue** marks the endpoint.

• **Standard Titration Procedure:**

1. *Standardization:* Titrate standard hard water (SHW) against EDTA solution $\rightarrow V_1$ mL.
2. *Total Hardness:* Titrate unboiled water sample against EDTA $\rightarrow V_2$ mL.
3. *Permanent Hardness:* Boil water sample, filter out precipitated carbonates, and titrate the filtrate against EDTA $\rightarrow V_3$ mL.

• **Formulas:**

- Let standard hard water concentration be S mg/mL of CaCO_3 .
- Volume of sample taken = V mL.
- 1 mL of EDTA = $\frac{V_{shw} \times S}{V_1}$ mg of CaCO_3 eq = z mg.
- Total Hardness = $\frac{V_2 \times z}{V} \times 1000$ ppm.
- Permanent Hardness = $\frac{V_3 \times z}{V} \times 1000$ ppm.
- Temporary Hardness = Total Hardness – Permanent Hardness.

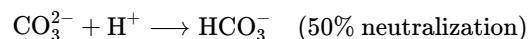
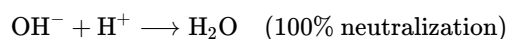
2. Estimation of Total Alkalinity

- **Principle:** Alkalinity is a measure of water's capacity to neutralize mineral acids, caused primarily by hydroxide (OH^-), carbonate (CO_3^{2-}), and bicarbonate (HCO_3^-) ions.
- **Mutual Exclusion Rule:** Because OH^- and HCO_3^- react immediately to form carbonate ($\text{OH}^- + \text{HCO}_3^- \rightarrow \text{CO}_3^{2-} + \text{H}_2\text{O}$), they **cannot coexist** in equilibrium. Possible combinations are:

1. Only OH^-
2. Only CO_3^{2-}
3. Only HCO_3^-
4. $\text{OH}^- + \text{CO}_3^{2-}$
5. $\text{CO}_3^{2-} + \text{HCO}_3^-$

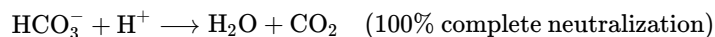
• **Two-Indicator Titration Scheme:**

- Sample is titrated with standard acid ($N/50 \text{ H}_2\text{SO}_4$).
- Step 1: **Phenolphthalein Indicator** (Endpoint at pH = 8.3, volume = P mL):



$$P = [\text{OH}^-] + \frac{1}{2}[\text{CO}_3^{2-}]$$

- Step 2: **Methyl Orange Indicator** (Endpoint at pH = 3.8, cumulative volume from start = M mL):



$$M = [\text{OH}^-] + [\text{CO}_3^{2-}] + [\text{HCO}_3^-]$$

• **Alkalinity Calculation Table:**

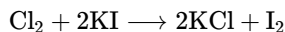
Relationship between P and M	Hydroxide (OH^-)	Carbonate (CO_3^{2-})	Bicarbonate (HCO_3^-)
$P = 0$	0	0	M
$P = M$	$P = M$	0	0
$P = \frac{1}{2}M$	0	$2P$	0
$P < \frac{1}{2}M$	0	$2P$	$M - 2P$
$P > \frac{1}{2}M$	$2P - M$	$2(M - P)$	0

3. Estimation of Free Chlorine

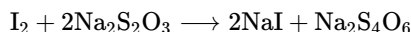
Chlorine is dosed into municipal supplies for disinfection. The remaining unreacted chlorine after demand is met is called **free residual chlorine** ($\text{HOCl} + \text{OCl}^-$).

- **Iodometric Titration Method:**

- Chlorine oxidizes iodide (I^-) quantitatively to elemental iodine (I_2) under acidic conditions (pH 3–4 using acetic acid):



- Liberated iodine is titrated against standard sodium thiosulfate solution (0.01 N $\text{Na}_2\text{S}_2\text{O}_3$):



- Freshly prepared **starch** indicator is added near the endpoint; the disappearance of the deep blue color marks the final titration reading:

$$\text{Free Chlorine (mg/L)} = \frac{V_{\text{thio}} \times N_{\text{thio}} \times 35.45 \times 1000}{V_{\text{sample}}}$$

- **DPD Colorimetric Method:** Free chlorine reacts instantly with *N*, *N*-diethyl-*p*-phenylenediamine (DPD) buffer at pH 6.3 to produce a magenta/red Würster dye, whose spectrophotometric absorbance is measured at 515 nm.

2.5 Solved Numericals

Numerical 1: Calculating Hardness from Water Composition

Problem: A water sample yields the following analytical results on chemical analysis:

- $\text{Ca}(\text{HCO}_3)_2 = 16.2 \text{ mg/L}$
- $\text{Mg}(\text{HCO}_3)_2 = 14.6 \text{ mg/L}$
- $\text{CaCl}_2 = 11.1 \text{ mg/L}$
- $\text{MgSO}_4 = 12.0 \text{ mg/L}$
- $\text{NaCl} = 58.5 \text{ mg/L}$
- $\text{SiO}_2 = 10.0 \text{ mg/L}$

Calculate the Temporary, Permanent, and Total hardness in ppm, °Fr, and °Cl.

Solution: Note: NaCl and SiO_2 do not contribute to water hardness.

$$\text{CaCO}_3 \text{ Equivalent} = \frac{\text{Mass (mg/L)}}{\text{Equivalent Weight of Salt}} \times 50$$

Constituent	Concentration (mg/L)	Equivalent Weight	Multiplication Factor	CaCO_3 Equivalent (mg/L or ppm)	Hardness Nature
$\text{Ca}(\text{HCO}_3)_2$	16.2	81	50/81	$16.2 \times \frac{50}{81} = 10.0$	Temporary
$\text{Mg}(\text{HCO}_3)_2$	14.6	73	50/73	$14.6 \times \frac{50}{73} = 10.0$	Temporary
CaCl_2	11.1	55.5	50/55.5	$11.1 \times \frac{50}{55.5} = 10.0$	Permanent
MgSO_4	12.0	60	50/60	$12.0 \times \frac{50}{60} = 10.0$	Permanent

- **Temporary Hardness:**

$$10.0 + 10.0 = 20.0 \text{ ppm}$$

- **Permanent Hardness:**

$$10.0 + 10.0 = 20.0 \text{ ppm}$$

- **Total Hardness:**

$$\text{Temporary} + \text{Permanent} = 20.0 + 20.0 = 40.0 \text{ ppm (or mg/L)}$$

- **Conversion to Other Units:**

- In Degree French (°Fr):

$$40.0 \times 0.1 = 4.0 \text{ }^{\circ}\text{Fr}$$

◦ In Degree Clark ($^{\circ}\text{Cl}$):

$$40.0 \times 0.07 = 2.8 \text{ }^{\circ}\text{Cl}$$

Numerical 2: EDTA Titration Calculation

Problem:

- 0.5 g of pure CaCO_3 was dissolved in dilute HCl and diluted to 1000 mL in a volumetric flask.
- 50 mL of this standard hard water required 40 mL of EDTA solution for titration.
- 50 mL of an unknown raw water sample required 25 mL of the same EDTA solution.
- 100 mL of the same raw water sample was boiled, filtered, and diluted back to 100 mL. A 50 mL aliquot required 10 mL of EDTA solution.

Calculate the Total, Permanent, and Temporary hardness of the water sample in ppm.

Solution:

Step 1: Standardization of EDTA Solution

- Concentration of SHW = $\frac{0.5 \text{ g}}{1000 \text{ mL}} = 0.5 \text{ mg/mL}$.
- Amount of CaCO_3 in 50 mL of SHW = $50 \times 0.5 = 25 \text{ mg}$.
- 40 mL of EDTA $\equiv$ 25 mg of CaCO_3 eq.
- 1 mL of EDTA = $\frac{25}{40} = 0.625 \text{ mg of CaCO}_3 \text{ eq.}$

Step 2: Total Hardness Calculation

- 50 mL of unknown water $\equiv$ 25 mL of EDTA = $25 \times 0.625 = 15.625 \text{ mg of CaCO}_3 \text{ eq.}$
- Hardness per liter (1000 mL):

$$\text{Total Hardness} = \frac{15.625}{50} \times 1000 = 312.5 \text{ ppm}$$

Step 3: Permanent Hardness Calculation

- 50 mL of boiled water $\equiv$ 10 mL of EDTA = $10 \times 0.625 = 6.25 \text{ mg of CaCO}_3 \text{ eq.}$
- Hardness per liter (1000 mL):

$$\text{Permanent Hardness} = \frac{6.25}{50} \times 1000 = 125.0 \text{ ppm}$$

Step 4: Temporary Hardness Calculation

$$\text{Temporary Hardness} = \text{Total Hardness} - \text{Permanent Hardness}$$

$$\text{Temporary Hardness} = 312.5 - 125.0 = 187.5 \text{ ppm}$$

Part 3: Solid Waste Management

3.1 Classification of Solid Waste

Solid waste encompasses unwanted, discarded non-liquid materials generated from domestic, agricultural, commercial, and industrial activities.

SOLID WASTE CLASSIFICATION

Waste Stream	Representative Constituents
Municipal Solid (MSW)	Food scraps, paper, cardboards, plastics, yard trimmings
Industrial Solid Waste	Fly ash, foundry sand, metallurgical slag, red mud
Hazardous Waste	Toxic heavy metals, spent organic solvents, batteries
Biomedical Waste	Sharps, infectious pathological tissues, used bandages
E-Waste	Discarded PCBs, cathode ray tubes, lithium-ion cells

Key Physical & Chemical Properties

- Moisture Content:** Critical parameter for biological processes:

$$M = \left(\frac{w - d}{w} \right) \times 100\%$$

where w is wet mass and d is oven-dry mass (105 °C).

- **Density:** Determines required collection vehicle capacity and landfill compaction ratios (loose MSW: 100–150 kg/m³; compacted landfill: 500–700 kg/m³).
- **Proximate Analysis:** Quantifies moisture, volatile combustible matter, fixed carbon, and non-combustible ash content.
- **Ultimate Analysis:** Determines elemental composition: percentage of Carbon (C), Hydrogen (H), Oxygen (O), Nitrogen (N), Sulfur (S), and residual ash.
- **Calorific (Heating) Value:** Higher Heating Value (HHV) evaluated via Bomb Calorimeter to establish suitability for thermal recovery (direct incineration requires > 5000 kJ/kg).

3.2 Integrated Waste Management Hierarchy

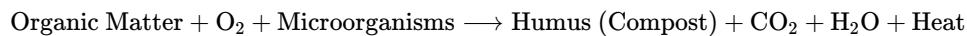
The internationally accepted waste management hierarchy prioritizes source reduction over final disposal:

1. **Source Reduction & Prevention:** Minimize waste production at the manufacturing stage through clean product design.
2. **Reuse:** Extend product life cycles directly without industrial reprocessing (e.g., refillable glass bottles, refurbished components).
3. **Recycling & Material Recovery:** Reprocess segregated waste streams into raw feedstocks (e.g., shredding PET bottles into polyester fiber).
4. **Energy Recovery:** Thermal or biological conversion of non-recyclable waste into heat, electricity, or synthetic fuels (e.g., anaerobic digestion, waste-to-energy incineration).
5. **Final Disposal:** Containment in engineered sanitary landfills with environmental monitoring.

3.3 Waste Treatment & Disposal Methods

1. Composting

Biological decomposition and stabilization of organic matter carried out by microbial populations under controlled thermophilic conditions:



- **Process Phases:**
 1. *Mesophilic Phase (20–40 °C):* Mesophilic bacteria and fungi break down simple soluble carbohydrates and sugars (1–2 days).
 2. *Thermophilic Phase (45–65 °C):* Spore-forming thermophilic bacteria (*Bacillus*, *Actinobacteria*) degrade proteins, fats, and complex hemicelluloses. Elevated temperatures destroy plant seeds and human pathogenic bacteria (e.g., *Salmonella*).
 3. *Cooling & Maturation Phase:* Temperature returns to ambient; fungi and actinomycetes humify complex lignins, forming dark, nutrient-rich soil humus.
- **Optimal Operating Conditions:**
 - **Carbon to Nitrogen (C:N) Ratio: 25:1 to 30:1** (High C slows the process; low C converts excess N into noxious ammonia gas, NH₃).
 - **Moisture Content: 50% to 60%** (Below 40% suppresses biological activity; above 65% floods pore spaces, creating foul anaerobic conditions).
 - **Aeration:** Minimum oxygen levels maintained above 10% via mechanical turning or forced aeration blowers.
 - **pH Range:** Neutral to slightly alkaline (6.5–8.5).
- **Methods:**
 - *Windrow Composting:* Organic waste is piled in long parallel rows (1.5–2.5 m tall) and turned mechanically at regular intervals.
 - *In-Vessel Composting:* High-rate, enclosed agitated drum systems featuring automated temperature, moisture, and airflow control.
 - *Vermicomposting:* Earthworms (*Eisenia fetida*) ingest organic waste alongside microorganisms, producing fine, nutrient-rich worm castings.

2. Sanitary Landfilling

Engineered burial of solid waste in designed cells isolated from surrounding groundwater and surface soil.

CROSS-SECTION OF AN ENGINEERED SANITARY LANDFILL

[Final Cap Layer]	Topsoil + Native Grass Veg Cover
-----	-----
[Barrier Cap]	Compacted Clay + HDPE Geomembrane

[Refuse Cells]	COMPACTED SOLID WASTE CELLS (Interspersed with daily 15 cm clean soil covers) ▲ ▲ ▲ ▲ ▲ (Vertical Methane Gas Wells)
[Leachate Drainage Bed]	Perforated HDPE Collector Pipes embedded in Gravel
[Composite Base Liner]	60-mil (1.5 mm) High-Density Polyethylene (HDPE)
[Clay Barrier]	>= 90 cm Compacted Low-Permeability Clay ($k < 10^{-7}$ cm/s)
[Subsoil Foundation]	Natural Native Geologic Strata

• Core Engineering Layers:

1. *Composite Impermeable Base Liner*: A minimum 90 cm thick layer of compacted clay (hydraulic conductivity $k \leq 10^{-7}$ cm/s), lined above by a thick, flexible 1.5–2.0 mm HDPE geomembrane.
2. *Leachate Collection & Removal System (LCRS)*: Perforated HDPE collector pipes bedded inside a 30 cm drainage layer of washed gravel sloped toward central sumps, where collected liquid is pumped out for wastewater treatment.
3. *Daily Soil Cover*: At the close of each working day, freshly dumped refuse is compacted and capped with 15 cm of soil to prevent rodent access, suppress odor, and shed rainwater.
4. *Landfill Gas Extraction*: Vertical perforated pipes extract landfill gas (50–60% CH₄, 40–50% CO₂) produced by anaerobic decomposition, routing it to a gas engine for electrical generation or to a flare stack.

3. Thermal Processes

THERMAL PROCESS COMPARISON				
Parameter	Incineration	Pyrolysis	Gasification	
Atmosphere	Excess O ₂ (120%)	Zero Oxygen	Starved O ₂ (20–40%)	
Temperature	850 - 1150 °C	400 - 800 °C	800 - 1200 °C	
Primary Output	Flue gas + Heat	Bio-oil + Char	Syngas (CO + H ₂)	
Volume Reduction	85 - 90%	70 - 80%	80 - 90%	

1. Incineration (Direct Controlled Combustion):

- High-temperature thermal conversion in excess oxygen (stoichiometric oxygen: 120–200%) inside moving-grate or fluidized-bed combustion chambers at temperatures between **850 °C and 1150 °C**.
- *Waste-to-Energy (WtE)*: Boiler tubes absorb combustion heat to generate high-pressure steam, driving a steam turbine generator.
- *Residues*: Bottom ash (80–90%, non-toxic mineral clinker used as construction aggregate) and toxic fly ash (10–20%, captured by baghouse fabric filters and electrostatic precipitators, treated as hazardous waste).
- *Flue Gas Treatment*: Lime slurry spray scrubbers neutralize acid gases (SO₂, HCl); activated carbon injection adsorbs vapor-phase dioxins and furans.

2. Pyrolysis:

- Thermal degradation of carbonaceous waste in the **complete absence of oxygen** at temperatures between **400 °C and 800 °C**.
- Endothermic process that cracks complex polymers into:
 - *Liquid Fraction (Bio-oil)*: Pyrolytic condensable hydrocarbons; can be upgraded to diesel-range liquid fuels.
 - *Gas Fraction (Syngas)*: Methane, carbon monoxide, and hydrogen.
 - *Solid Residue (Char)*: Fixed carbon and ash; usable as low-grade activated carbon or solid fuel.

3. Gasification:

- Partial oxidation of carbonaceous solid material at **800–1200 °C** under a **sub-stoichiometric oxygen atmosphere** (20–40% of theoretical air).
- Produces synthetic fuel gas (**Syngas**: predominantly CO + H₂), which can be cleaned and burned in combined-cycle gas turbines or converted via Fischer-Tropsch synthesis into synthetic chemicals.

4. Recycling and Reuse



Gathering	(NIR, Float)	(Clean Imp)	(Compounded)	Products
-----------	--------------	-------------	--------------	----------

- **Plastics:** Mechanical recycling shreds and sorts clean thermoplastic items by polymer grade (PET #1, HDPE #2, PP #5), re-extruding them into commercial pellets. Chemical recycling (depolymerization) uses glycolysis, methanolysis, or pyrolysis to convert mixed plastics back into base chemical monomers.
- **Metals:** Ferrous scrap is separated using cross-belt electromagnets; non-ferrous aluminum is isolated using eddy-current separators. Remelting recycled aluminum requires only 5% of the energy needed to refine bauxite ore via the Hall-Héroult process.
- **Paper:** Repulped in warm water baths to separate cellulose fibers, washed, de-inked via surfactant froth flotation, bleached with hydrogen peroxide, and rolled into recycled kraft board.
- **Glass (Cullet):** Sorted by color (clear, amber, green), crushed into uniform cullet, and introduced directly into industrial melting furnaces. Adding cullet lowers furnace melting temperatures by 200–300 °C, reducing operating fuel requirements.

Sources

1. [Engineering Chemistry: Polymers Overview | PDF - Scribd](#)

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